Critical Points, Extrema & Graphing

Name _____

Date _____ Hour _____

PROCEDURE

1. Critical points. Find $f'(x)$ and factor. Critical points are x-values in the domain where $f'(c) = 0$ or $f'(c)$ is undefined.

2. Second Derivative Test. $f''(c) > 0$: local min. $f''(c) < 0$: local max. $f''(c) = 0$ or undefined: inconclusive.

3. First Derivative Test. Check the sign of f' on each side of c. + to −: local max. − to +: local min. No change: neither.

4. Global extrema. On a closed interval, compare f at every critical point and both endpoints. On all real numbers, think about end behavior.

The interactive flow chart is in the Calculus I Supplemental Materials: Local & Global Extrema, Flow Charts.

PROBLEM 1 $f(x) = x^3 - 3x^2 - 9x + 5$ on the interval $[-2, 6]$

(a) Find $f'(x)$ and factor it completely.

(b) Find the critical points.

(c) Find $f''(x)$, then use the **Second Derivative Test** on each critical point.

$f''(x) =$ _____

critical point c	f″(c)	f″(c) positive or negative?	local max, local min, or inconclusive

(d) Write each local extremum as a point (x, y).

Local max: _____ Local min: _____

(e) Find the **global** maximum and minimum of f on $[-2, 6]$. Fill in the table with every candidate.

x					
f(x)					

Global max: _____ at x = _____ (endpoint) Global min: _____ at x = _____

Problem 2: When the Second Derivative Test Fails

PROBLEM 2 $f(x) = x^6 - 6x^4$ on all real numbers

(a) Find $f'(x)$ and factor it completely. (Hint: pull out the greatest common factor first, then look for a difference of squares.)

(b) Find the critical points.

(c) Find $f''(x)$, then use the **Second Derivative Test**. Write INCONCLUSIVE if it cannot decide.

$f''(x) =$ _____

critical point c	$f''(c)$	result of the Second Derivative Test

(d) Use the **First Derivative Test** to settle every critical point. Complete the sign chart for $f'(x)$.



critical / possible x							
test value x							
sign of $f'(x)$							
f is							

(e) Classify each critical point and give its coordinates.

$x = -2$: _____

$x = 0$: _____

$x = 2$: _____

(f) Does f have a global maximum? A global minimum? Explain using end behavior and your values from (e).

Problem 3: A Cusp

PROBLEM 3 $f(x) = x^{2/3}(x - 5)$ on the interval $[-1, 8]$

(a) Rewrite f as a sum, $f(x) = x^{5/3} - 5x^{2/3}$. Then find $f'(x)$ and write it as a single fraction with the common factor pulled out.

(b) Find **all** critical points in $[-1, 8]$. (Check where $f'(x) = 0$ *and* where $f'(x)$ is undefined but f is defined.)

(c) Here is $f''(x) = \dfrac{10(x + 1)}{9x^{4/3}}$. Can the **Second Derivative Test** be used at each critical point? What does it say?

critical point c	f″(c)	what the Second Derivative Test says

(d) Use the **First Derivative Test** to classify every critical point. Complete the sign chart for $f'(x)$.



critical / possible x					
test value x					
sign of f′(x)					
f is					

(e) Give the coordinates of each local extremum. Round decimals to two places.

x = 0: _____

x = 2: _____

(f) Find the global maximum and minimum of f on $[-1, 8]$. Fill in every candidate. Is the local minimum also the global minimum?

x					
f(x)					

Global max: _____ at x = _____ Global min: _____ at x = _____

Problem 4: Putting It All Together

PROBLEM 4 $f(x) = 3x^5 - 5x^3$ on all real numbers

You will use every tool from Problems 1 to 3, and also concavity. This page is the analysis. Page 5 is the graph.

(a) Critical points. Find $f'(x)$, factor completely, and list the critical points.

(b) Increasing and decreasing. Complete the sign chart for $f'(x)$.

critical / possible x							
test value x							
sign of $f'(x)$							
f is							

(c) Local extrema. Classify each critical point. $f''(x) = 60x^3 - 30x$. Use it to check -1 and 1, and explain what happens at 0.

(d) Concavity and inflection points. Factor $f''(x) = 60x^3 - 30x$, find where it equals 0, and complete the sign chart for $f''(x)$.

$f''(x) =$ _____ zeros: $x =$ _____

critical / possible x							
test value x							
sign of $f''(x)$							
f is							

(e) List the inflection points as coordinates (exact and decimal).

(f) Intercepts and global extrema. Find the y-intercept and any x-intercepts (factor f). Does f have global extrema? Why?

Problem 4, Continued: The Graph

Graph $f(x) = 3x^5 - 5x^3$ using your work from page 4. Plot and label every local extremum, inflection point, and intercept. Show concavity. The graph leaves the window near the top and bottom, so draw arrowheads where it goes off.



Key points. List them from left to right.

point (x, y)	what it is	point (x, y)	what it is

Reflect. At x = 0 the Second Derivative Test told you nothing. In one or two sentences, what did the First Derivative Test tell you, and what does that look like on your graph?